

Customer Behaviour Analysis

Internship Project Report

Prepared By

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Submitted To

Alfido Tech

Tools Used

- Python
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
-

Dataset

Customer Behaviour Analysis Dataset (Kaggle)

Submission Year

2026

1. Objective

The objective of this project is to analyse customer purchasing behaviour using transaction data. The analysis focuses on identifying purchasing patterns, customer segments, and business opportunities through data analysis techniques.

The project aims to:

- Understand customer purchasing behaviour.
- Clean and prepare the dataset for analysis.
- Create meaningful features from the available data.
- Perform Exploratory Data Analysis (EDA).
- Segment customers using RFM Analysis.
- Generate business insights and recommendations.
- Support better customer engagement strategies.

2. Dataset Description

The dataset contains customer transaction records collected from an e-commerce business.

Dataset Summary

Attribute	Value
Total Transactions	250,000
Total Customers	Approximately 50,000
Features	13 Original Columns
Missing Values	Returns column only
Duplicate Records	None

Important Columns

- Customer ID
- Purchase Date
- Product Category

- Product Price
- Quantity
- Total Purchase Amount
- Payment Method
- Returns
- Age
- Gender
- Churn

```
[3]: df.head()
```

Out[3]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	31	Female	0
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	31	Female	0
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	31	Female	0
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	0.0	John Rivera	31	Female	0
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	1.0	John Rivera	31	Female	0

Figure 1: Preview of the Customer Transaction Dataset

3. Data Cleaning

Before analysis, the dataset was cleaned to improve data quality.

The following operations were performed:

- Checked missing values.
- Filled missing values in the Returns column with 0.
- Verified duplicate records.
- Converted Purchase Date into Date Time format.
- Removed duplicate columns (Customer Age).
- Verified final data types.

After cleaning, the dataset contained no missing values in the analysed columns.

```
[7]: df.isnull().sum()
Out[7]: Customer ID      0
Purchase Date      0
Product Category    0
Product Price      0
Quantity           0
Total Purchase Amount 0
Payment Method      0
Customer Age        0
Returns            47382
Customer Name       0
Age                0
Gender              0
Churn               0
dtype: int64
```

Figure 2: Dataset Information After Data Cleaning

4. Feature Engineering

Several new features were created from the Purchase Date column.

These include:

- Purchase Year
- Purchase Month
- Purchase Weekday
- Age Group

These additional features helped identify customer purchasing trends across time and different age categories.

5. Exploratory Data Analysis

Several visualizations were created to understand customer behaviour.

The analysis included:

- Product Category Distribution
- Revenue by Product Category
- Payment Method Distribution
- Revenue by Age Group
- Monthly Revenue Trend
- Gender Distribution
- Correlation Heatmap
- Purchase Amount Distribution

These visualizations revealed purchasing trends, customer preferences, and revenue patterns.

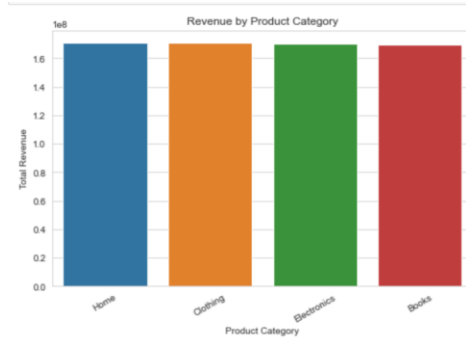


Figure 3: Revenue by Product Category



Figure 4: Monthly Revenue Trend

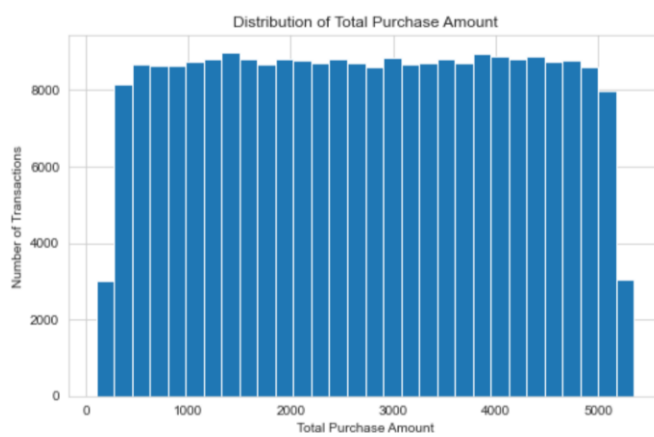


Figure 5: Distribution of Total Purchase Amount

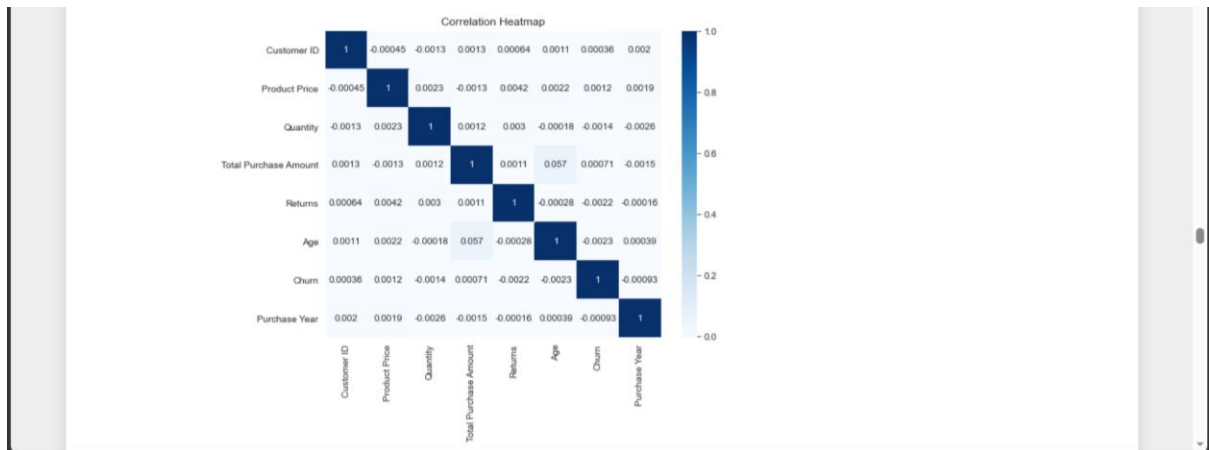


Figure 6: Correlation Between Numerical Variables

6. Customer Segmentation (RFM)

RFM Analysis was used to classify customers based on:

Recency

How recently a customer purchased.

Frequency

How often a customer purchases.

Monetary

How much money the customer spends.

Customers were grouped into the following segments:

- Champions
- Loyal Customers
- Potential Loyalists
- Regular Customers
- Frequent Buyers
- Big Spenders
- At Risk Customers
- Lost Customers

This segmentation helps businesses personalize marketing strategies and improve customer retention.

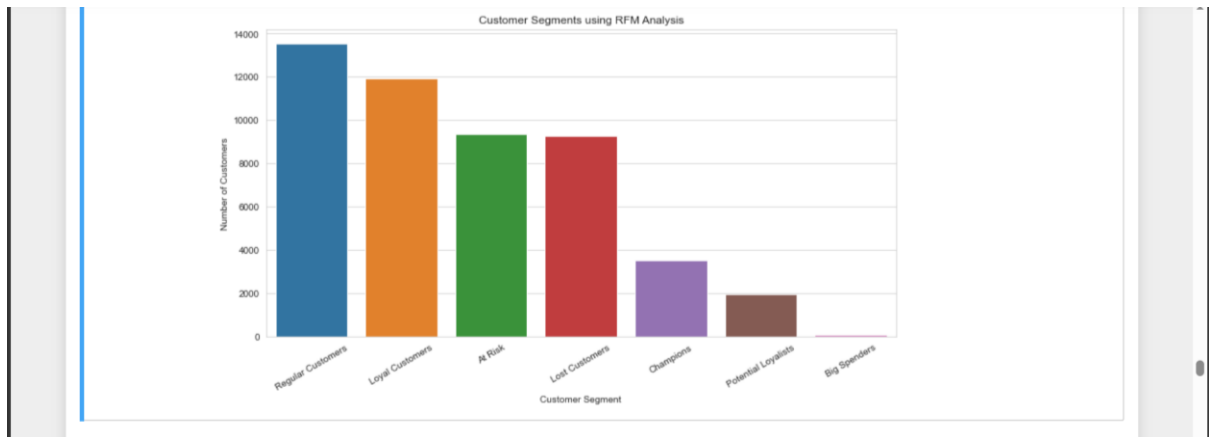


Figure 7: Customer Segments using RFM Analysis

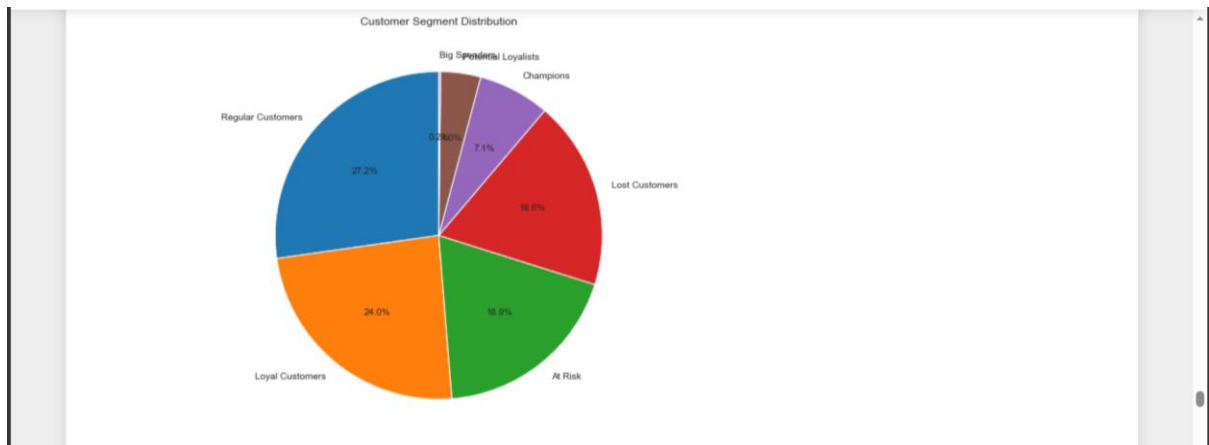


Figure 8: Percentage Distribution of Customer Segments

	Recency	Frequency	Monetary
Segment			
At Risk	345.95	6.44	17572.06
Big Spenders	261.94	4.87	19132.89
Champions	35.40	8.43	24147.47
Lost Customers	649.95	2.91	7950.54
Loyal Customers	91.74	6.52	17369.03
Potential Loyalists	86.48	4.44	15243.64
Regular Customers	170.86	3.41	8806.82

Table 1: Average RFM Values by Customer Segment

7. Key Findings

- The dataset contains 250,000 customer transactions.
- Customers purchase products from multiple categories.
- Different payment methods are preferred by different customers.
- Monthly sales indicate seasonal purchasing trends.
- RFM Analysis successfully identified valuable customer segments.
- Regular Customers represent the largest customer group.
- Champions contribute significantly to business value.
- At Risk and Lost Customers require re-engagement.

8. Recommendations

Based on the analysis, the following recommendations are provided:

1. Reward Champions with exclusive loyalty benefits.
2. Send personalized offers to Loyal Customers.
3. Re-engage At Risk Customers through targeted campaigns.
4. Promote high-performing product categories.
5. Use customer segmentation to personalize product recommendations.

9. Conclusion

The project successfully analysed customer purchasing behaviour using Python and data visualization techniques.

Data cleaning, feature engineering, exploratory data analysis, and RFM Analysis provided valuable insights into customer behaviour and purchasing patterns.

The findings can help Alfido Tech improve customer engagement, increase customer retention, and support data-driven business decisions.

10. Tools & Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Thank You

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